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„A small Catalogue of Dust Clouds
within 1.25 kpc of the Sun“

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Abstract

This L^AT_EX template provides example on how to format and display text, mathematical formulas, and insert tables or images. There is a lot more you can do with L^AT_EX, for more information check out <https://en.wikibooks.org/wiki/LaTeX>.

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2.2. The 3D dust map

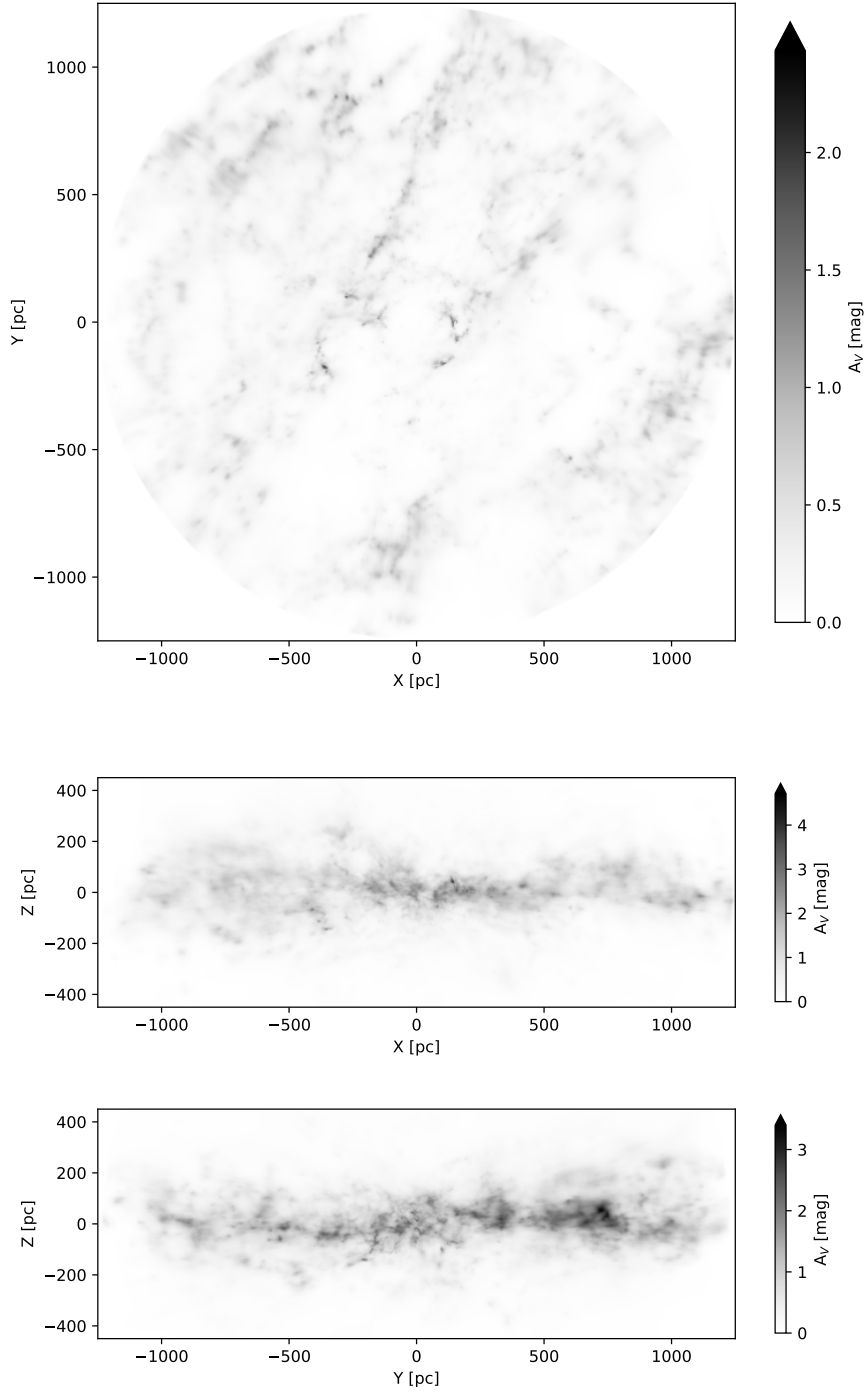


Figure 2.1.: Projections of the Dataset presented by (Edenhofer et al., 2024), in a volume of $2.5 \text{ kpc} \times 2.5 \text{ kpc} \times 900 \text{ pc}$, along the Z-, Y-, and X-axis respectively.

3. Methods

3.1. Cataloguing

To explore, catalogue, and categorize the data, the Python Project `glueviz` was used (Beaumont et al., 2015; Robitaille et al., 2019). The Clouds in the Data set were selected by a process of visual inspection, where visually interesting structures, and those that could be easily separated from the Background were naturally favoured.

3.2. Analysis

3.2.1. Size and Location

For each cloud object, a cutoff density n_C was determined, where the main body of the cloud being considered is contiguous, and it is separated from surrounding high-density regions. Due to the nature of the visualization software used to create the 3D plots in this work, the `plotly` python package, (Inc, 2015), two variables determine the cutoff density: f_{min}, f_{max} . The cutoff density n_C is:

$$n_C = \frac{f_{min} + f_{max}}{2}. \quad (3.1)$$

See Appendix A for a table of the parameters for each cloud. The values for f_{min} and f_{max} are presented in unitless Extinction E , as in the ZGR catalogue.

In Fig. 3.1a, Cloud 4 from the catalogue is shown as an example. The scale on the right side is a colorscale for the surrounding diffuse dust, whereas the cloud's internal structure is not shown, merely it's shell. Within the green-marked region, all dust is considered to belong to the cloud and contributes to the Mass Calculation.

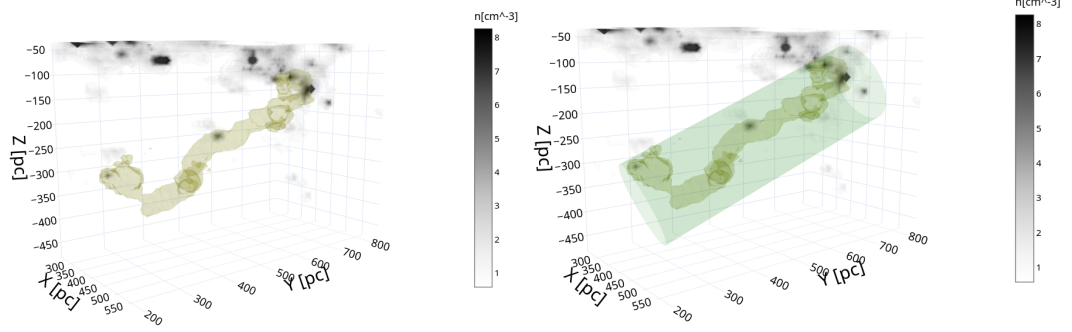
After the density threshold was set, a cylinder was placed to encapsulate the cloud and closely approximate the filament. The center of this cylinder is used to describe the location of the cloud in heliocentric Cartesian coordinates. The radius and height of the cylinder are in Parsec. See Fig.3.1b for a visualisation of a cylinder around the same example catalogue member. [interactive – FiXme](#)

Owing to the rotational symmetry of cylinders around their central axis, only two angles are necessary to describe their location, α as the rotation about the z -axis, and β as the inclination to the galactic plane. In the Heliocentric Cartesian System, when looking on the galactic plane from above, as in Fig.2.1a, the galactic centre is to the right. The connection line between the sun and the galactic centre corresponds to $\alpha = 0^\circ$. After a counter-clockwise rotation by α , the cylinder is inclined by β .

The parameters used to define and encode the cylinders are summarized in Tab. 3.1.

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3. Methods



(a) The boundary of Cloud 4 in this Catalogue (b) A cylinder has been defined to include the cloud boundaries

Figure 3.1.: The green region in the middle represents Cloud 4 of the Catalogue within the surrounding dust. While the internal structure of the diffuse dust around it is displayed, for the Catalogue member itself only it's boundary is shown, here roughly at $n_C = 0.58 \text{ cm}^{-3}$.

Variable	Meaning
$r[\text{pc}]$	Radius of the Cylinder
$h[\text{pc}]$	Height of the Cylinder
$\vec{c}[\text{pc}]$	The Center of the Cylinder as a vector in Cartesian coordinates
$\alpha[\text{rad}]$	Rotation about the z -axis
$\beta[\text{rad}]$	Inclination to the galactic plane
N_r, N_h, N_θ	Amount of slices in the r, h, θ -axes
f_{min}, f_{max}	Used to encode the threshold density, and for visualization

Table 3.1.: The variables used to encode the characteristics of the cylinders, also the inputs for the Mass Calculation Algorithm.

3.2.2. Mass Calculation

A Conversion of the Dust Extinction measured by (Edenhofer et al., 2023) to a Mass of the Clouds was done by a process adapted from (O'Neill et al., 2024).

To convert the Extinction map of (Edenhofer et al., 2023) to a mass in the clouds, the cylinders set up around the clouds were utilized. Following (O'Neill et al., 2024) in converting the the Extinction to a density with

$$n_i = \frac{1}{(A_G/N_H)} \frac{dA_{G,i}}{1 \text{ pc}} \quad (3.2)$$

$$= 1653 \text{ cm}^{-3} A'_{ZGR,i}, \quad (3.3)$$

where $A'_{ZGR,i}$ is the differential Extinction in the i -th slice of the volume in unitless ZGR23 extinction, see (Zhang et al., 2023a) for the definition, and n_i the resulting density in a volume element. $A_G/N_H \simeq 4 \times 10^{-22} \text{ cm}^2 \text{ mag}$ is used, as in (Zucker et al., 2021), using (Draine, 2003, 2009).

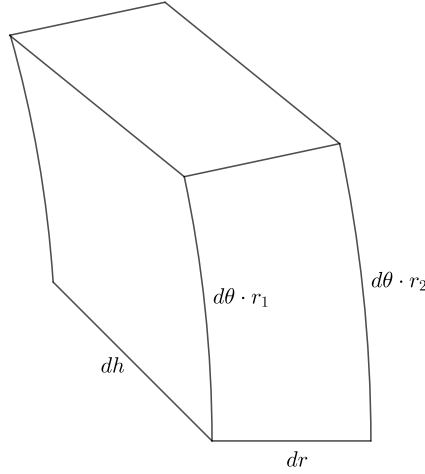


Figure 3.2.: An Element of the Cylinder.

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An illustration of a volume element can be seen in Fig. 3.2. It's volume is derived, beginning with the area in the r - θ -plane, as part of an annulus:

$$da = (r_{i+1}^2 - r_i^2) \frac{d\theta}{2}. \quad (3.4)$$

Since the annulus is separated into n_θ parts, $d\theta$ is $\frac{2\pi}{n_\theta}$. Thus we divide the area of the whole annulus by n_θ , and arrive at the above equation. Multiplying by dh and utilizing $r_i = i \cdot dr$, we get:

$$dv_i = dh \frac{d\theta}{2} (dr)^2 ((i+1)^2 - i^2) \quad (3.5)$$

$$= \left(i + \frac{1}{2}\right) d\theta dh (dr)^2. \quad (3.6)$$

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We now sum over all elements:

$$M = 1.37m_p \sum_{i,j} n_{i,j} dv_{i,j}, \quad (3.7)$$

iterating by i over the rings, and by j over all elements within that ring, only where $n_i \geq n_C$, the threshold density. Here, m_p is the Proton mass, and the factor 1.37 is calculated from abundances of Helium and heavier molecules contributing to the total mass.

3.2.3. Mass Uncertainty Estimation

Statistical

The statistical uncertainty of the masses is mostly dependent on the standard deviation of the $E+$ Extinction map and thus n , see (3.7), but the calculation of dv_i is assumed to be containing statistical uncertainties as well, where the choice of $dh, d\theta$ and dr is error-prone, with an uncertainty of its own value square-rooted. Thus, the combined uncertainty is:

$$\Delta M_{stat} = 1.37m_p \sqrt{\left(\sum_i \Delta n dv_i\right)^2 + \left(\sum_i n \Delta dv_i\right)^2}. \quad (3.8)$$

The uncertainty of the density in the cylinder elements is also provided in the $E+$ dataset and can again be sampled using the `dustmaps` package. For all Clouds, the statistical uncertainty is less than 0.5 % of their mass.

Systematic

The systematic uncertainty inherent in the selection process of the clouds, as well as the parameter, lead to a much greater uncertainty than the statistical uncertainty would predict. To estimate the uncertainty in the mass calculation, the input variables into the mass calculation algorithm were varied.

Due to computational restraints, it was not possible to vary all parameters, as the mass calculation algorithm needed to sample the $E+$ -dataset using the `dustmaps` framework, which took multiple seconds for each run.

To determine which parameters have the highest influence on the result, Cloud 4 was used as a representative example of the catalogue, as there were regions of higher density, which would influence the calculated mass, close to the structure itself, but not filaments of such mass that the definition of the cloud boundaries are questionable.

Varying all input parameters for Cloud 4, the ones that influenced the total estimated mass of the cloud the most, were selected to be varied in the Uncertainty estimation of all clouds. These most influential parameters can be seen in Tab.3.2. Also, multiple different values for $d\theta, dh, dr$ were tested, but had very little influence when varied in the order of magnitude of a few parsecs.

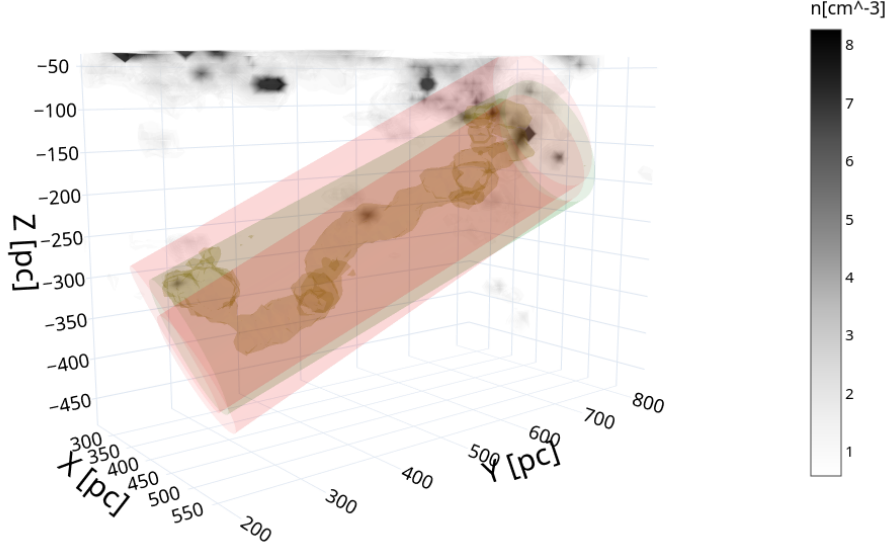


Figure 3.3.: Cloud 4 of this catalogue, as in Fig.3.1b, with 2 erroneous cylinders of the uncertainty estimation process layered on top.

See Fig.3.3 for a visualisation of the same cloud in Fig.3.1, with two example cylinders that were used for the alternative mass calculation.

Variable	Variation
r	$\pm 10\%$
α	$\pm \frac{0.25}{18} \cdot \pi$
β	$\pm \frac{0.25}{18} \cdot \pi$
t	± 20 pc in each direction(x, y, z)
f_{max}	$\pm 10\%$

Table 3.2.: A description of the varied parameters, and the magnitude of that variation.

Varying over all variables in the defined range, 2187 masses were calculated for each cloud. The mean of these samples are considered as the masses for the regions. The systematic uncertainty is then the standard deviation within these results for a specific cloud. For the total uncertainty, the statistical and systematic uncertainty were added together.

3.3. Code

The Code used in this project will be available in a github repository of the Author. [link](#) ! – FiXme

3. Methods

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The main files in this repository are `FuncDef.py` and `ErrorCalc.py`. An Example of a Plotting Script is also provided.

FuncDef

A module holding all of the most used functions, like querying the `dustmaps` package in specific points, or defining a `Cloud` class, or calculating Masses.

ErrorCalc

A script to calculate the Systematic Uncertainty of the Masses by looping over iterations of input parameter variations for each cloud.

4. Results

A catalogue of 23 clouds has been identified. The respective parameters that define the clouds, according to Tab.3.1, can be found in Appendix A. In Fig.4.1, the distribution of the clouds over a low-resolution sampling of the Extinction map can be seen. The images are not projections, but views from the perspective of an observer along the axes into the cloud distribution in 3D. [interactive version online](#), [fix consistent look of plots](#)

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4.1. Overall

As an overview of the catalogue members, Fig. 4.2 shows a histogram of the masses of the Clouds in this catalogue. The clouds with their respective Masses are also represented in Tab.4.1. As a visualisation, the percentage error is plotted against the mass of the clouds in Fig.4.4. The high uncertainty at low masses is not a systematic error in the Mass and Uncertainty calculation, but is a result of those clouds being in very high-density environments, where very small variations in the input parameters lead to great differences in the calculated mass. The same can be said for Cloud 3, which is the mark very close to $M = 2 \times 10^5 M_{\odot}$.

4.2. Discussion of individual objects

4.2.1. Vela Complex

Due to the Nature of this work, very few steps were taken to include specific regions. Owing to its prominence and isolation in the Edenhofer dustmap, the Dust accumulation in the Vela region was included in this catalogue as Cloud 19. This work in no way intends to determine conditions for membership to any cloud group.

Both the volume of the encompassing cylinder and the determined mass veer far from the other results in this work, which focuses much more on smaller and more diffuse structures. As a result, both its Volume and Mass eclipse the respective second largest catalogue member by a factor of 3, as can be seen in Figures 4.2 and 4.3.

4.2.2. Multiple-Armed objects

Two Regions could not be fit into a singular cylinder, or were more closely approximated by using multiple cylinders. We will shortly consider them here.

4. Results

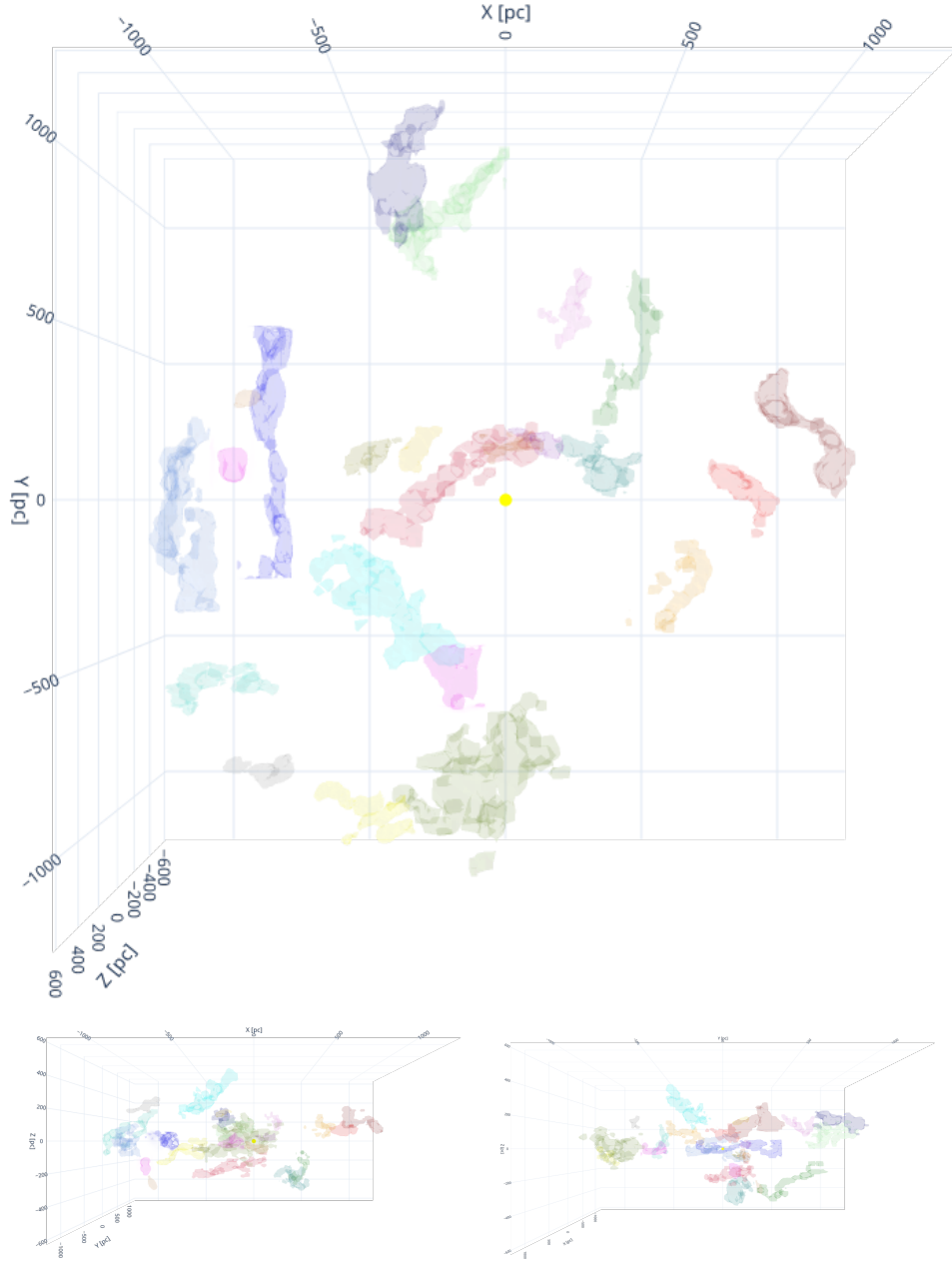


Figure 4.1.: A visualisation of the distribution of the clouds in Heliocentric Cartesian Space.

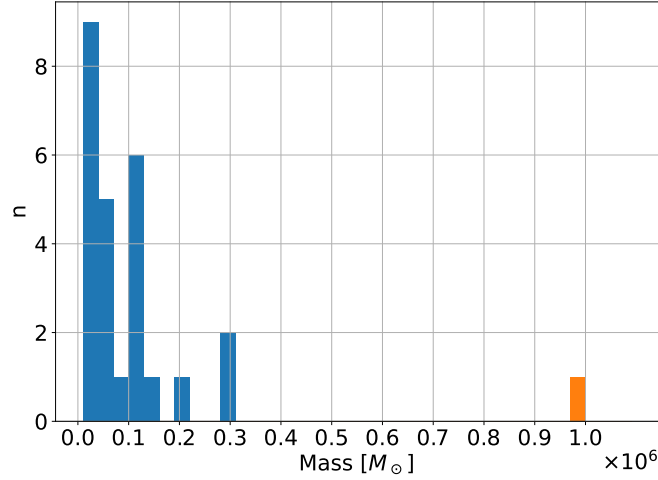


Figure 4.2.: A histogram of the Masses of the 23 Catalogue members. The object with the highlighted Mass is Cloud 19, further discussed in 4.2.1.

Cloud 8

Cloud 8 is remarkable due to the difference in inclination between the different cylinders. While the first two named arms are roughly parallel, the third arm, 8c, is much differently inclined. The inclination β differ from $\sim -10^\circ$ for arms a, b and $\sim 41^\circ$ for arm c . In Fig.4.5, two of the arms are shown with the cylinders around it. The difference in Inclination is very noticeable. Again, see Appdx.A for details.

Due to its surroundings and the closeness to the massive object approximated in Cloud 19 of this catalogue, the uncertainty of the individual masses of the arms is very high.

Cloud 21

A different case was found in Cloud 21, where the environment was dense with high-density objects, and no singular cylinder was found that encompassed the contiguous object that was considered, that didn't also include a lot of the surrounding dust. The surroundings of the cylinders is also highly relevant in the Uncertainty of the Mass Calculation, thus leading to very high uncertainties for these arms, as can be seen in Tab.4.1.

4.2.3. Other remarkable Objects

Cloud 5

A very diffuse region, with the lowest cutoff density of all regions. Cloud 5 seems to be an arm above the galactic plane, where very low-density regions connect nexuses of

4. Results

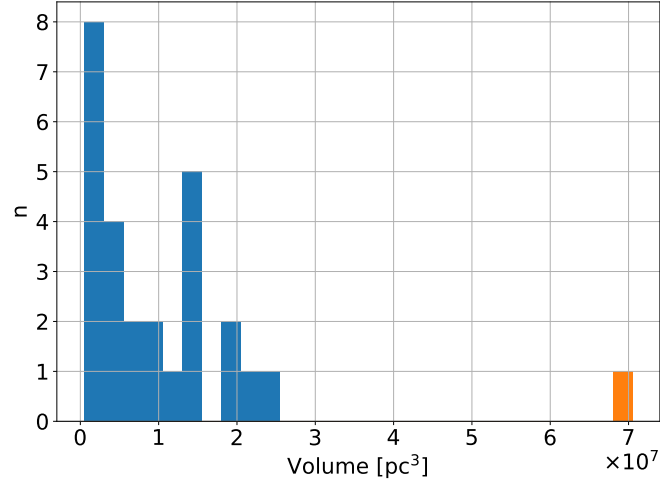


Figure 4.3.: The Volumes of the 23 Catalogue members in a Histogram. As in Fig.4.2, the orange cloud is discussed in 4.2.1.

cloud cores that are, in comparison to other internal cores in this catalogue, of very low density.

With the second largest volume of all cylinders in the set, it has the 5th lowest mass with $M_5 = (4.27 \pm 0.39) \times 10^4 M_\odot$.

Cloud 10

The smallest and, with a mass of $M_{10} = (1.077 \pm 0.043) \times 10^4 M_\odot$, the least massive Cloud in the Catalogue is Object Nr. 10. It is also remarkable due to a very high distance to the galactic plane, as it lies roughly 400 pc below it.

It is also a very isolated higher-density region, and there is very little other dust in the vicinity. The closest other high concentration of dust is also classified in this catalogue, in Cloud 16.

Cloud 14

Cloud 14 is a relatively high-mass region, with $M_{14} = (3.07 \pm 0.24) \times 10^5 M_\odot$ actually the second biggest contiguous cloud. It is hard to separate from cloud 9, however, which lies very close. In the cylinder of cloud 14, a small part of 9 is present as well, skewing the Mass Estimate towards higher results.

With its centre around $z = 250$ pc, it is a massive cloud region that lies high above the galactic plane. More intentional study will have to be conducted to understand the dynamic processes that lead to a massive contiguous dust cloud like this to be observed at such a high altitude.

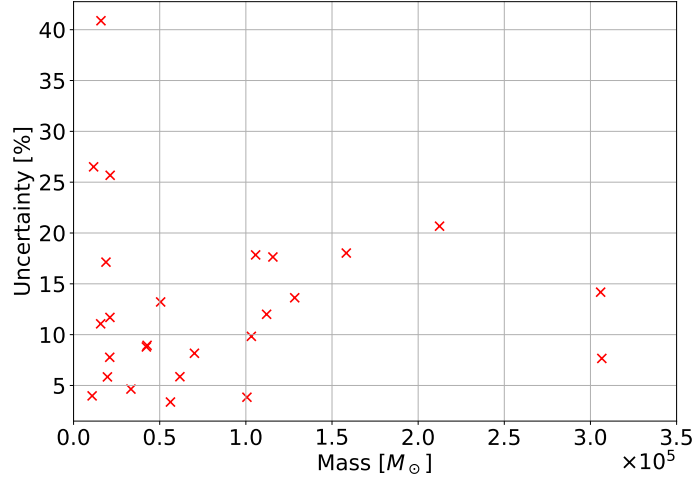


Figure 4.4.: The uncertainty in the masses plotted against the masses of the clouds. Cloud 19 is missing from this visualisation.

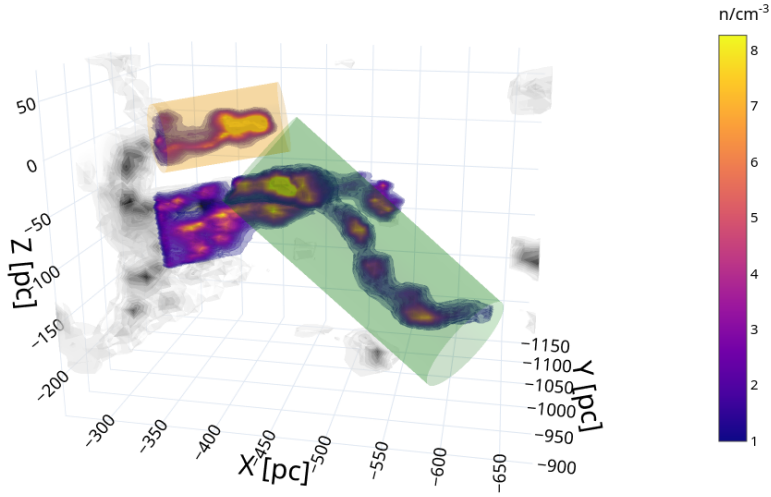


Figure 4.5.: A visualisation of Cloud 8a, b and c, including their inner structure. Around Arm a, an orange cylinder has been drawn, and around arm c, a green cylinder.

4. Results

Cloud Nr.	Mass in $10^3 M_\odot$	$\Delta M/M[\%]$
1	100.6 $\pm$ 3.9	3.8
2	61.7 $\pm$ 3.7	5.9
3	212 $\pm$ 44	20.7
4	50.5 $\pm$ 6.7	13.2
5	42.7 $\pm$ 3.9	8.9
6	103 $\pm$ 11	9.8
7	19.6 $\pm$ 1.2	5.8
8a	11.6 $\pm$ 3.1	26.5
8b	21.2 $\pm$ 5.5	25.7
8c	18.9 $\pm$ 3.3	17.1
9	306 $\pm$ 44	14.2
10	10.77 $\pm$ 0.43	4.0
11	21.0 $\pm$ 1.7	7.8
12	70.2 $\pm$ 5.8	8.2
13	42.3 $\pm$ 3.8	8.8
14	307 $\pm$ 24	7.7
15	33.3 $\pm$ 1.6	4.6
16	56.2 $\pm$ 1.9	3.4
17	158 $\pm$ 29	18.0
18	15.7 $\pm$ 1.8	11.1
19	989 $\pm$ 53	5.3
20	106 $\pm$ 19	17.8
21a	128 $\pm$ 18	13.6
21b	112 $\pm$ 14	12.0
21c	116 $\pm$ 21	17.6
22	21.1 $\pm$ 2.5	11.7
23	15.9 $\pm$ 6.6	40.9

Table 4.1.: The Masses of the Clouds

5. Discussion

The definitions of the Clouds, their boundaries, and their Masses all suffer significantly due to the high amount of manual work and non-standardized decision making that has gone into each step in the process. Since this catalogue was not based on necessarily previously discussed regions and clouds, a comparison to previous works by other authors is difficult, but relevant for a few elements, revealing the discrepancies between this catalogue’s results and theirs.

5.1. Reliability of Mass Estimates

Considering the circumstances of this study, the uncertainties in the mass estimates are still very small. Owing to the small uncertainties in differential extinction in the $E+$ dataset, and thus small uncertainties in the density of the volume elements, the actual measurements are very accurate, with a statistical uncertainty of $\sim 0.2\%$ across the board.

The reliability of the measurements and the uncertainty measurements is thus mostly based on the quality of the parameters chosen to define the clouds, which is addressed in 3.2.3, and the actual calculation method.

The mass values presented in this work are based on the same presumptions that plague the datasets used to create this catalogue, including, but not limited to:

- $R_V = 3.1$
- There exists a universal extinction curve, as presented by (Zhang et al., 2023a).

A big issue regarding the quality of the calculation is that only spatial distribution of the dust clouds was considered, and the regions being contiguous in velocity space and belonging to the same dynamical group did not factor into defining the clouds or their boundaries. However, the order of magnitude for the masses of the clouds are very accurate, regardless of the method used to obtain their values.

5.2. Comparison to previous Studies, where applicable

Again, previous catalogues and definitions of dust clouds has not been considered, which is why a direct comparison of mass estimates of the clouds defined here to results of other works is very difficult.

A few separate comparisons arise, however. Cloud 12 from this catalogue is the dust part of the California Molecular Cloud. Lombardi et al. (2010) use wide-field extinction

5. Discussion

maps to calculate the total mass of the California Cloud total to be $M = 132 \times 10^3 M_\odot$, with $43 \times 10^3 M_\odot$ at $A_K \geq 0.2$. Since they use integrated Extinction along the LOS, a direct comparison between extinction thresholds is hard, but as this work presents $M_{12} = (70.2 \pm 5.8) \times 10^3 M_\odot$, the cutoff for cloud membership in this catalogue seems to be somewhere between $A_K = 0.1$ and $A_K = 0.2$.

Cloud 22 and 23, in total amassing $\simeq 38 \times 10^3 M_\odot$, represent small parts of the Local Chimney presented in O’Neill et al. (2024). The total mass of the shell reported is $M_{tot} = (6.0 \pm 0.7) \times 10^5 M_\odot$. These results appear consistent, as huge parts of the region presented by O’Neill et al. are not considered in the mass calculation in this work.

The Vela complex, further separated into multiple smaller Molecular Clouds, among others, in (Soler et al., 2017), is presented to have a total mass of about $M = 5 \times 10^5 M_\odot$. Since there has been no further conditions on Membership to Cloud 19 apart from spatial distribution, and a density cutoff, a harsh overestimation of the masses of $M_{19} = (9.9 \pm 0.5) \times 10^5 M_\odot$ seems reasonably agreeable.

5.3. Future Improvements

The $E+$ dataset has huge potential in defining dust cloud catalogues in the local galactic environment. Major Improvements upon this work can be achieved by standardising the methods in this work, especially in the processes of selecting the objects, and setting the density thresholds. A Master’s Thesis addressing this issue is being worked on.

Also, cross-matching with existent dust cloud catalogues will be necessary to better define Catalogue Membership, utilise a consistent naming convention, and obtain comparable data.

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A. Parameters for the Clouds

Tab.A.1 presents the parameters that have been input into the mass calculation algorithm, according to Tab.3.1. The parameter f_{min} is set for all clouds, $f_{min} = 1 \times 10^{-6}$.

The unit of f_{max} is unitless extinction, as is the $E+$ dataset itself, and is thus used as the direct limiter for the mass calculation algorithm, before the unitless extinction is converted to a density according to (3.3). [link to csv – FiXme](#)

! – FiXme

Cloud Nr.	r [pc]	h [pc]	α [°]	β [°]	c_x [pc]	c_y [pc]	c_z [pc]	$f_{max}[10^{-3}E]$
1	80	300	-45	0	740	40	100	1.6
2	110	345	60	-10	530	-270	110	1.6
3	90	800	-90	0	-775	150	0	1.6
4	90	550	78	24	425	510	-260	0.7
5	120	550	-37.5	40	-340	-320	370	0.2
6	95	500	10	40	-910	-595	80	1.6
7	65	250	7.5	30	-750	-825	230	1
8a	35	130	10	-10	-360	-1075	0	1
8b	40	260	25	-5	-440	-1050	-70	2
8c	50	250	1	41	-520	-980	-90	1
9	120	430	45	0	-140	920	120	2
10	60	170	30	-60	-910	350	-380	0.5
11	60	185	50	10	-260	150	240	0.9
12	60	220	22	-2.5	-430	150	-70	5
13	100	300	-27	-15	320	130	-290	0.4
14	100	480	80	-10	-320	1010	250	1.6
15	110	200	90	0	-170	-570	-20	0.4
16	100	120	0	0	-920	120	-215	0.8
17	100	700	35	10	-180	50	-250	1
18	105	325	80	-35	195	600	230	0.6
19	200	550	45	0	-45	-875	0	2
20	100	420	-45	-10	960	200	190	0.6
21a	70	340	84	5	-980	-190	0	1.6
21b	55	420	84	5	-1040	-40	80	1.6
21c	85	460	87	-11.5	-1070	40	-60	1.6
22	55	200	8	-5	20	170	-60	1
23	40	200	-17	5	80	190	40	0.8

Table A.1.: The parameters of the clouds in the catalogue.